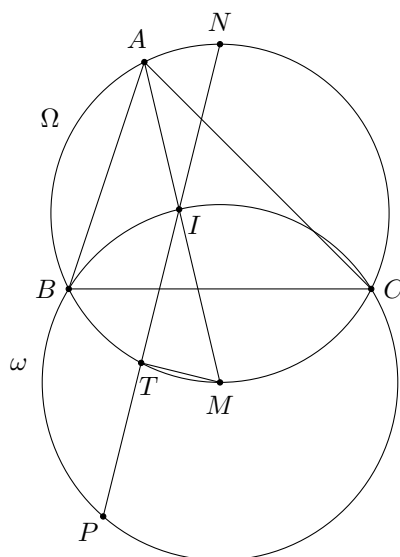


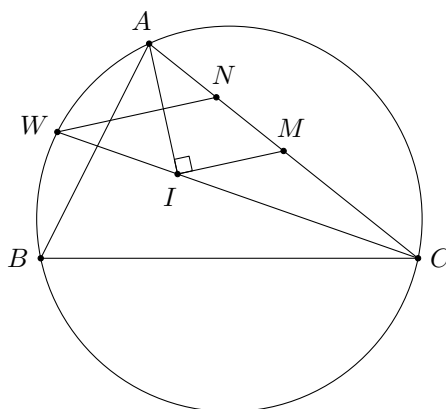
Problem 53. In triangle ABC , let I be the incentre. Denote the circumcircles of $\triangle ABC$ and $\triangle BIC$ by Ω and ω respectively. In Ω , let M be the midpoint of arc BC not containing A , and let N be the midpoint of arc BC containing A . Suppose NI meets ω the second time at P . Prove that the midpoint of IP lies on Ω .



Solution. As M is the centre of ω , $MI = MP$. Therefore, if we let T be the midpoint of IP , $\angle MTN = \angle MTI = 90^\circ$. We are done because MN is a diameter of Ω .

Remark: T is exactly the point of tangency of the A -mixtilinear incircle and the circumcircle. □

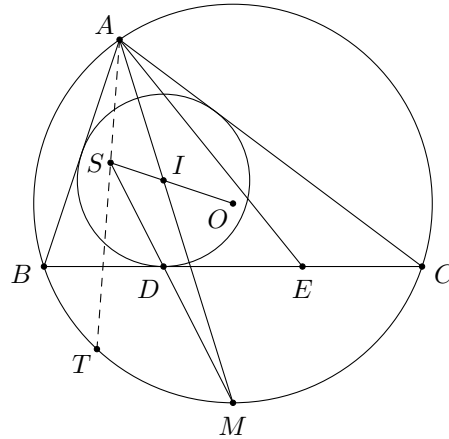
Problem 54 (Sharygin Geometry Olympiad Final 2017 Grade 9 Problem 2). Let I be the incenter of a triangle ABC , M be the midpoint of AC , and W be the midpoint of arc AB of the circumcircle not containing C . It is known that $\angle AIM = 90^\circ$. Find the ratio $CI : IW$.



Solution. Let ℓ be the line passing through W perpendicular to AI . Then as $WA = WI$, ℓ passes through the midpoint of AI . As $\ell \perp AI$ and $\angle AIM = 90^\circ$, $\ell \parallel IM$, hence ℓ also passes through N , the midpoint of AM . Now $\ell \parallel IM$ also implies that $\triangle CIM \sim \triangle CWN$, and $CI : IW = CM : MN = 2 : 1$.

Remark: The condition implies that the A -mixtilinear incircle ω touches AC at M . If ω touches AB and the circumcircle at P and T , then W, P, T are collinear, and the shooting lemma implies $WA^2 = WP \cdot WT$. Therefore, ℓ is exactly the radical axis of point circle A and ω , giving that ℓ passes through the midpoint of AM . □

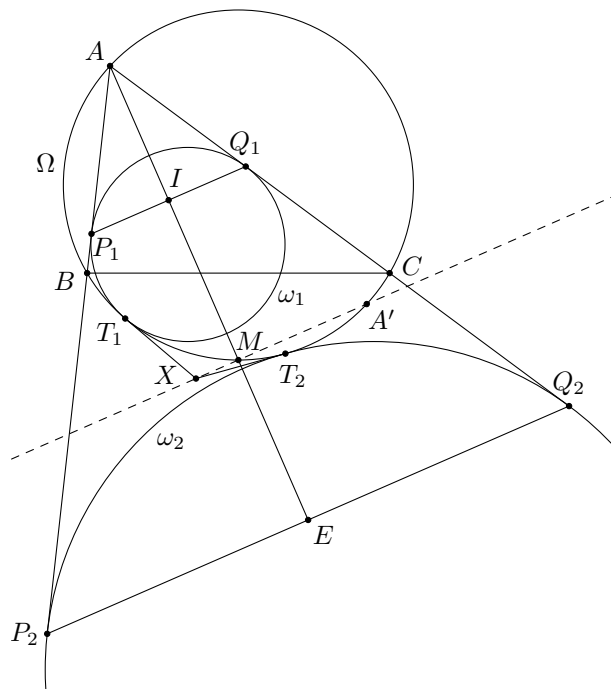
Problem 55. In triangle ABC , let I and O be the incentre and circumcentre respectively. Suppose the incircle and A -excircle touch BC at D and E respectively. Let M be the second intersection of AI and the circumcircle of $\triangle ABC$, and let S be the intersection of DM and OI . Prove that $\angle BAS = \angle EAC$.



Solution. Since the tangent at D to the incircle is parallel to the tangent at M to the circumcircle, the external centre of similitude between the incircle and the circumcircle lies on DM . This centre also lies on OI , and so it is the point S . Let the A -mixtilinear incircle touch the circumcircle of $\triangle ABC$ at T . Then we already know that $\angle BAT = \angle EAC$, so what remains is to show that A, S, T are collinear, which is immediate by Monge's theorem on circumcircle, incircle and A -mixtilinear incircle.

Remark: Let U be the touching point of the B -mixtilinear incircle and the circumcircle, and let V be the touching point of the C -mixtilinear incircle and the circumcircle. By symmetry, AT, BU, CV are concurrent at S . By the result of this problem, this intersection point is the isogonal conjugate of the Nagel point. $\square$

Problem 56 (Taiwan TST 2015 Round 3 Quiz 1 Problem 2). Let Ω be the circumcircle of the triangle ABC . Two circles ω_1, ω_2 are tangent to each of the circle Ω and the rays AB, AC , with ω_1 interior to Ω , ω_2 exterior to Ω . The common tangent of Ω, ω_1 and the common tangent of Ω, ω_2 intersect at the point X . Let M be the midpoint of the arc BC (not containing the point A) on the circle Ω , and the segment AA' be the diameter of Ω . Prove that X, M , and A' are collinear.



Solution. Note that the two common tangents are radical axes, therefore X is the radical centre of Ω, ω_1 and ω_2 . Now let ℓ be the radical axis of ω_1 and ω_2 , hence $X \in \ell$.

Suppose ω_1 touches AB and AC at P_1 and Q_1 , and ω_2 touches AB and AC at P_2 and Q_2 . Then the incentre I and the A -excentre E are the midpoints of P_1Q_1 and P_2Q_2 respectively, with $P_1Q_1 \parallel P_2Q_2$. Therefore, as ℓ passes through the midpoints of P_1P_2 and Q_1Q_2 , it also passes through the midpoint of IE , which is exactly M .

It remains to notice that $\ell \parallel P_1Q_1 \perp AM$ and $\angle AMA' = 90^\circ$, so that A' also lies on ℓ as desired. $\square$